

Advances in Deep Learning: A Survey of Modern Architectures

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Abstract

Machine learning algorithms fall into supervised, unsupervised, and reinforcement learning paradigms. Supervised learning requires labeled data where inputs pair with output labels. The algorithm learns a mapping generalizing to unseen data. Common tasks include classification, regression, and sequence labeling. Data quality and quantity significantly impact performance. CNNs exploit spatial locality through convolutional filters sliding across input. Each filter detects specific features like edges or textures. Pooling reduces spatial dimensions while preserving important features. Modern architectures like VGG, ResNet, and EfficientNet use depth, skip connections, and neural architecture search respectively. Regularization prevents overfitting via constraints or penalties. L1 (Lasso) encourages sparsity. L2 (Ridge) prevents large weights. Dropout randomly deactivates neurons during training, forcing redundant representations. Batch normalization stabilizes training by normalizing layer inputs across mini-batches. Data augmentation artificially expands training sets. Gradient descent is foundational for neural network optimization. SGD approximates gradients using mini-batches, introducing noise that helps escape local minima. Adam, RMSprop, and Adagrad adapt learning rates per parameter based on historical gradient information. Learning rate scheduling and warmup strategies further improve convergence.

1 Introduction

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2 Related Work

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Gradient descent is foundational for neural network optimization. SGD approximates gradients using mini-batches, introducing noise that helps escape local minima. Adam, RMSprop, and Adagrad adapt learning rates per parameter based on historical gradient information. Learning rate scheduling and warmup strategies further improve convergence. Regularization prevents overfitting via constraints or penalties. L1 (Lasso) encourages sparsity. L2 (Ridge) prevents large weights. Dropout randomly deactivates neurons during training, forcing redundant representations. Batch normalization stabilizes training by normalizing layer inputs across mini-batches. Data augmentation artificially expands training sets. Machine learning algorithms fall into supervised, unsupervised, and reinforcement learning paradigms. Supervised learning requires labeled data where inputs pair with output labels. The algorithm learns a mapping generalizing to unseen data. Common tasks include classification, regression, and sequence labeling. Data quality and quantity significantly impact performance.

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3 Method

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3.1 Core Formulation

Regularization prevents overfitting via constraints or penalties. L1 (Lasso) encourages sparsity. L2 (Ridge) prevents large weights. Dropout randomly deactivates neurons during training, forcing redundant representations. Batch normalization stabilizes training by normalizing layer inputs across mini-batches. Data augmentation artificially expands training sets. Machine learning algorithms fall into supervised, unsupervised, and reinforcement learning paradigms. Supervised learning requires labeled data where inputs pair with output labels. The algorithm learns a mapping generalizing to unseen data. Common tasks include classification, regression, and sequence labeling. Data quality and quantity significantly impact performance. Gradient descent is foundational for neural network optimization. SGD approximates gradients using mini-batches, introducing noise that helps escape local minima. Adam, RMSprop, and Adagrad adapt learning rates per parameter based on historical gradient information. Learning rate scheduling and warmup strategies further improve convergence.

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3.2 Implementation Details

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4 Experiments

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Table 1: Benchmark comparison

Method	Dataset	Acc (%)	F1
Baseline	ImageNet	72.3	0.71
ResNet-50	ImageNet	76.1	0.75
ResNet-152	ImageNet	78.3	0.77
Ours	ImageNet	79.1	0.78

4.1 Datasets and Setup

Machine learning algorithms fall into supervised, unsupervised, and reinforcement learning paradigms. Supervised learning requires labeled data where inputs pair with output labels. The algorithm learns a mapping generalizing to unseen data. Common tasks include classification, regression, and sequence labeling. Data quality and quantity significantly impact performance. CNNs exploit spatial locality through convolutional filters sliding across input. Each filter detects specific features like edges or textures. Pooling reduces spatial dimensions while preserving important features. Modern architectures like VGG, ResNet, and EfficientNet use depth, skip connections, and neural architecture search respectively. Transfer learning leverages knowledge from large source datasets to improve small target datasets. Feature extraction freezes the pre-trained model and trains only a new classifier. Fine-tuning unfreezes some layers at low learning rates. This avoids overfitting while benefiting from rich representations learned on large-scale data.

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4.1 Datasets and Setup (cont.)

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4.2 Main Results

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